

Let m be a fixed positive integer and N a positive integer that will later go to ∞ . Let M be a symmetric $\binom{N}{m}$ -by- $\binom{N}{m}$ matrix. The rows and columns are indexed by the subsets of $[N] = \{1, \dots, N\}$ that have exactly m elements. (Indeed there are obviously $\binom{N}{m}$ many of them.)

Let S and T be two subsets of $[N]$ with m elements. Finally, let $f_N(l) : [m] \rightarrow \mathbb{N}$ be a function with N as parameter. The matrix M satisfies the following:

If $|S \cap T| = l$, then the entry at row S and column T has value $f_N(l)$.

We want to do asymptotic analysis on the following integral by letting $N \rightarrow \infty$:

$$\binom{N}{m}^{-1} \log \int_{[0, \infty)^{\binom{N}{m}}} \exp(x^T M x) dx. \quad (1)$$

The setup may seem confusing at the first. To give some intuition, consider the special case that $m = 1$. Then we have a normal N -by- N matrix that can be written as

$$M = a_N \cdot Id + b_N \mathbb{1}^T \mathbb{1}, \quad (2)$$

where $\mathbb{1}$ is the vector with all entries 1. (1) can be written as

$$\frac{1}{N} \log \int_{[0, \infty)^N} \exp(b_N \|x\|_1^2 + a_N \|x\|_2^2) dx. \quad (3)$$

The above expression can be written as the expectation of a function of $\|X\|_1$, where $X = (|X_1|, \dots, |X_N|)$ and X_i are i.i.d standard Gaussian. We can use Laplace method and compute the asymptotics.

The problem becomes drastically more difficult when $m > 1$. Take $m = 2$ for instance, (2) becomes

$$M = a_N \cdot Id + b_N \mathbb{1}^T \mathbb{1} + c_N M', \quad (4)$$

where M' satisfies that the entry at row $\{a, b\}$ and column $\{c, d\}$ is 1 if $\{a, b\} \cap \{c, d\} = 1$ and 0 otherwise. ($m = 2$ means the matrix is indexed by 2-subsets.) We have

$$x^T M x = a_N \|x\|_2^2 + b_N \|x\|_1^2 + c_N x^T M' x. \quad (5)$$

Unless we know the rate function of $\langle X, M' X \rangle$ then the integral in (1) can be handled by simply applying the contraction principle. However, it is hard to compute the rate function of $\langle X, M' X \rangle$, and we don't know if it is possible.

Questions: 1. For case $m = 2$, is it possible to derive a LDP for $\langle X, M' X \rangle$? If so, can this method be extended to any fixed positive integer m ?

2. If the setup is too cumbersome. Then for general symmetric, positive-definite matrix M , do we have a systematic approach to compute (1)?

Remark 1. The matrix M is actually the inverse of a covariance matrix. I managed to compute the inverse of the covariance, but what waits for me after a road block is another road block.